GPW Evolution '98, version 0.2

What changed since 0.1, and why

A plain-language tour, by Ferrim

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TL;DR

Version 0.1 gave GPW an SC and a smarter AI. Version 0.2 is mostly about making the rest of the race hold together around them:

- The SC behaves better: it stops teleporting, the queue forms in time, and the blue-flag investigations that made no sense under it are gone.
- The AI handles rain much better. It re-plans when the track changes, splits its choices between team-mates, never pushes a stop it had already planned further away, and the weather itself now moves with the clock instead of the leader's lap.
- A car that comes into the pits for a repair now uses the stop: the wing gets fixed for real, tyres and fuel are changed, and the rest of its stops are planned again from there.
- Parts wear out where they should. The wear table has been redone, Monaco is as brutal as it deserves, and the AI eases off its driving orders as a part runs out instead of driving it into the ground.
- The wind is a real thing now, tyre grip keeps paying off beyond the middle of its scale, and the FIA regulations move the pace of the whole field.
- Penalties are applied where they used to fall through: at the end of the race, to lapped cars, and in the pits. And a car that has crossed the line after the flag no longer loses its result if it breaks down.
- Three new screens: **Live Championship Standings** (drivers and constructors, with arrows for who gains and loses places), **Car Internals** and a much bigger **Position** tab.
- A game saved in the middle of a race now remembers the pit stop times.

Most of this is described with what the game does and why. As in the first narrative, there are no figures: they are the numbers the game works with.

How to read this document

Each chapter covers one area, and you can read them in any order. If you haven't read the first narrative, the short version is that I keep what makes GPW feel like GPW and fix what is plainly broken before adding anything new. This one only covers what is new since 0.1.

There are no percentages, seconds, lap counts or money in the text, for the same reason as before: those are the numbers the game works with, and they have changed while I tested.

Many of the changes here came from you: from the form, the forum and the Discord. Where that is the case I say so. The names are the ones people chose to appear with.

What to expect

The new features have been through a test bench and through real races, and I'm happy with how they behave, but you will find the cases I haven't. If you do, the generator behind the EXE can switch off any single feature, so a bad idea can be taken out without losing the rest.

The SC, round two

The first version worked, but anyone who played with it knew the weak spots, and several of them were reported here.

The ghost car and the queue

The SC is represented by a ghost car that leaves the pit lane, gets caught by the leader, and goes back in at the end. In 0.1 the ghost sometimes seemed to jump backwards just before it came in, and in some races a few cars were left half a lap behind when the queue should have been formed. Two things were wrong. The ghost kept its own pace while the leader braked on the last lap, so it ran away from him until the game decided it was too far and put it back; and the cars that were far from the one in front weren't closing up fast enough.

Now the ghost brakes together with the leader on the last lap (just a bit less), the limit before the game moves it back is wider, and cars that are far behind close up quicker. The distances between the cars while it is out, and the gap at the restart, are a bit smaller than in 0.1.

Blue flags

A lapped driver sitting between two leaders behind the SC was being investigated for ignoring the blue flags. Under the SC nobody can overtake, so nobody can be blamed for not letting anyone by. That check is off while the SC is out. Blue flag messages may still pop up, and when there are lapped cars behind the SC the first few laps after it can still bring penalties, so maybe in a later release I'll lower the odds for the first couple of laps after the SC.

What I haven't fixed

There is a report that the ghost sometimes misses the pit entry and stays out on track. What I changed attacks the jump, not what happens afterwards. I don't know whether it is gone, and it probably gets worse at the higher game speeds.

Weather, and what the AI does about it

The weather runs on the clock

In 0.1, qualifying and the race handled the weather in different ways: in qualifying it depended on the session time, and in the race on the leader's lap. That meant the track always changed at the same place, when the leader crossed the line, and you could wait for it. Now the race weather also runs on time, as qualifying does. It will still change, but not at a moment you can predict.

Re-planning when the water changes

This is the part of 0.2 that took the most rounds, and most of the rounds were fixes to my own work.

In 0.1 the AI re-planned when the track went wet or dry, but not when it went from wet enough for intermediates to wet enough for wets, or the other way round. Everybody stayed out, whatever tyre they were on, although they would pick the right one at their next scheduled stop. Now, when that happens, each AI team splits its choices: the first driver goes for the tyre that suits the new state of the track, and his team-mate gets the same chance if the first one doesn't. That idea came from the community (Acescheil).

Then it needed refining as it ran in real races:

- The first stop after a re-plan comes sooner, so a car that is on the wrong tyre doesn't carry on for several laps.
- A re-plan never delays a stop the car had already planned. This came from a car that had a stop coming up and ended up stopping later because the weather changed.
- The ideal tyre is the same for every driver, whatever his wet-weather skill: intermediates for damp, wets for wet. A driver who complains that he wants a different one doesn't change what is best.
- A minimum stint also applies after a re-plan, so a car no longer ends up with a stint that is absurdly short at the end.

The AI in the pits

A stop for a repair is a stop

When an AI car came in with a broken front wing in 0.1, the pit crew never got the order to fix it: the car stood still, left, and often retired soon afterwards. Now it is fixed for real. The same goes for any other breakdown, with a twist that came from the community (Acescheil): a car that is running in the points takes the risk and stays out, while one that is running down the field comes in to have it checked.

But a repair stop is also a stop: the car is in the pits anyway, so it used to be a waste not to change its tyres or take on fuel. Now the AI treats the unscheduled stop as the first stop of a new plan. It refuels, changes tyres, and re-plans the rest of its stops from the laps that are left. Whether it makes sense for it to finish with one stop fewer than it had planned depends on the race, the same way the weather re-plan decides.

The repair itself can still fail. That is how the original game was built, and I haven't touched it.

The pit stop bug is back in the fix list

A few of you noticed that in the pit lane a car with a low number could overtake one with a high number, as in the old pit stop bug. It turned out that the fix had been in the early test builds and dropped out of the package by mistake, so 0.1 was released without it. It is back. It is still an experiment: the results so far are encouraging but I can't prove it with the number of pairs I have seen.

The pit lane, circuit by circuit

Australia, Brazil, Spain and Austria now have a pit lane value of their own, because compared with the real stops of the period a stop was costing too much in some of them and too little in others. The rest keep the general value until I get to them.

Wear, failures and driving orders

The general wear table

Brazil was always hard on cars. After 0.1, Germany and Belgium became just as hard, to the point that a small team could not reach the end. The cause was the way I had scaled wear by the distance of the race: on those circuits the scaling stacked on top of an already high rate. The table of rates has been redone, with the scaling built in, and the closing of the session uses its own rates now too. The effect is that Germany and Belgium go back to something a small team can survive, Brazil is a bit kinder, and the circuits that used to wear very little wear a bit more.

Monaco

I had asked myself why a race at Monaco ended with almost everybody still running. Parts could only fail when a counter reached zero, and none did. Monaco now wears the parts of the cars more, and a healthy car has a small chance of hitting the barriers at the end of each lap. That includes your own cars. The SC treats it as an accident.

The AI and its driving orders

In 0.1 the AI only started easing off its orders when a part was almost gone. Now it does it in steps: as the counter of a part falls, the target order falls with it, one step at a time, for brakes, suspension, clutch, gearbox, hydraulics and throttle. A target is never lowered more than one step per lap, even when several parts ask for it at once. Cars live closer to the edge, which was the point: a car with a brake problem was pitting again and again late in the race.

Illegal aids

The AI is much less reckless about fitting illegal driving aids, and the chance falls as the season goes on.

Parts and reliability

The starting counter for reliability was the same for the two highest ratings in every part. That isn't the case any more. The design department had some similar slips: the progress counter of a technology part wasn't reset when the part was finished or at the end of the season, so the next one came out already complete; the CAD building did nothing; and the four stages of the chassis design weighed in the opposite order to how long they take. All of that has been fixed.

Pace, overtaking and the car

Overtaking

The speed bonus in an overtake used to depend on a ratio that nobody understood. Now the faster car gets a bonus that depends on how much faster it is than the other, with a limit, and the Overtaking value of the circuit counts only for the attacker, which makes Monaco what it should be: before, it was added to both the attacker and the defender, so the rating of the track made no difference. The effect of that value depends on the circuit, and I'm still not happy with some of them.

The wind

The wind existed in the game, but what it did was close to the opposite of what you would expect: the more wind you set up for, the more you lost, and with some settings nothing at all. Now the penalty depends on how far the wind setting of the car is below what the real wind asks for. A car with enough wind setting loses nothing; a car with none loses a clearly noticeable amount in a strong wind. This is a first version, and I'll take another look at it once more of you have played with it.

Tyres

Tyre grip made the car better up to the middle of its scale and then did nothing, while still costing fuel and tyre temperature. It keeps giving something now.

The FIA

The FIA regulations come up every year in the game, but they almost never changed anything. Three things have changed. The caps for engines, fuel and tyres are lower, so the regulations actually act on the suppliers. When a chassis regulation is approved with continuity, the design now starts by itself with the progress that came with it; before, pressing the button lost it. And the type of the regulations moves the base pace of the circuits: when nothing changes the cars get faster as the teams keep developing, and with a change they get slower.

Chassis and set-up

I have adjusted the starting chassis of five teams to my own handling table. And the set-up reserve is capped: you could earn more points than the game was meant to hold by testing, applying and taking them back.

Penalties and the final order

- A stop-go that had been given but not served by the end of the race now counts in the final order.
- If a penalty is enough to drop a car behind cars with fewer laps that finished behind him, he now drops behind them and counts as lapped. The Race Order panel shows it: how many laps down, and the penalty in red after the name.
- If you have a stop-go to serve, you can now call the pits freely during the first laps after it, to refuel and change tyres. When the time to serve it comes, the car comes in by itself, as before. Before, "Pit now" was blocked for the whole period.
- A car that has already crossed the line after the flag no longer loses its result if something breaks before the race is officially over. In 0.1 it was counted as retired, and its place and points went to the car behind.
- And, as I said above, there are no investigations for blue flags under the SC.

Qualifying

When it started to rain in a qualifying session, the whole AI left the pits at the same time, and a driver coming back after a bad lap could end up waiting in the pits and not run again. The AI leaves at spread-out intervals now, and a car that is outside the 107% can go out again. The state of the track when the pole is set is also kept at that moment: there was an edge case, when the weather changed around the time the pole was set, in which the AI thought the pole had been set on a drier track than it really was, and kept sending its cars out pointlessly.

The paddock and the money

- **Salaries:** the cap on what a team can pay, based on its income, now also applies to department chiefs, and there is a second pass at the end of the season.
- **Test drivers:** the fixed veto of the original game, which kept three drivers from ever being signed as test drivers, is gone. The signing news also says whether a driver has been signed as a test driver or as a race driver, with wordings and headlines of their own.
- **Pay drivers:** because of the bug with GPWEdit 1.1, the game was paying them instead of being paid by them. That is fixed for new games; a saved game keeps its own data.
- **Repairs** use the skill of the mechanical staff, not the commercial staff. That came from Jose21crisis.
- **Morale:** the highest levels of a department's morale gave too much extra effort. They give less now, and the lower ones are unchanged.
- **Sponsors:** in the commercial sponsors page, the contracts of more than one year with a supplier were shown as works deals when they were not. They now show the level that was signed. As far as I can tell it was a display problem, not a data one.
- **Audio:** in the game, Takagi was called Rosset and the other way round. The voice files are swapped now, so each driver is called by his own name.

New screens

Live Championship Standings

A new panel in the race screen, reached with the arrows of the panel after Car Internals. It has two columns: the drivers' championship and the constructors'. They show the table as it would be if the race ended at that moment: the points of the season plus the points of the position each driver is holding now, without counting the cars that have retired. Each row has its place, the name and the points. Your cars and your team are in yellow.

Next to the place there is an arrow: green and up if the driver has gained places compared with the championship before the race, red and down if he has lost them. It only appears if somebody has really been passed, so two drivers with no points don't move even if others score.

The arrows are drawn directly and not with the game's own text, so if they cause trouble on your system the generator can leave them out and keep the rest.

It shows all the drivers of the race, and it doesn't count a penalty that is applied after the flag.

Car Internals

For each of your drivers, a bar for each of the seven large parts of the car (brakes, clutch, electronics, gearbox, hydraulics, suspension and throttle), showing how much is left of each.

The Position tab

It used to show two cars ahead and two behind. It shows five and five now, with the window sliding so that the panel is always full.

Race Order at the start

At lap zero it showed an order that looked random. It shows the grid now.

Pit Stops

The panel shows the lap and time of each stop of every driver, and, as described below, keeps them when you save the game.

Saved games

If you saved a game in the middle of a race, the times of the pit stops came back as zero when you loaded it. The saved game now carries them, together with the other data that the new parts of the game use during a race. Games saved with 0.1 still load, with those figures at zero. An SC that is on track at the moment of saving is not kept.

What comes next

Some things are still on my list: the wind, the pace at Germany, a few circuits that still use a general pit lane value, the minimum stint of the last stint of a race, and the French and German versions of the new texts, which are not in the package. The ideas list is on the page of the project, where you can see it and add your own: GPW Evolution '98 · Send a proposal | Enviar una propuesta

Reporting bugs

If something in here doesn't feel right to you, or if you find a bug I haven't noticed, I'd like to hear about it, with the circuit, the round and what you were doing: on the forums at GPW General | GPM Planet, at the forum's Discord, or wherever you find me. Many of the changes in this version started there.

Happy gaming!